

Evolving methods for all-season farming

Not having accurate knowledge of climatic conditions in an area results in low-yield and low-quality crops. Devastated crops result in higher prices for consumers in the market. For enhancing the yield of crops without worrying about climatic conditions in a place, new ways of farming have come up. Automated agriculture has made indoor agriculture possible, thereby preventing land deterioration and increasing production of reliable food supply.

The concept of vertical farming is being adopted at a rapid pace. Vertical farms cultivate crops and raise cattle inside dedicated or mixed-use skyscrapers, which meet the growing requirements of urbanisation. This way plants can grow without sunlight. As sensitivity of crops to light varies based on wavelength, energy-efficient lighting using LEDs can ensure that light levels are optimised. Green Sense Farms has developed a commercial farm that utilises LED lights that are

customised for each crop. Advantages of such a system include efficient crop production irrespective of weather, so that consumers get fresh food throughout the year at low rates.

Smart greenhouse-based farming is one method where such environmental parameters as pressure, humidity, temperature and light levels are monitored and controlled with the help of IoT-based sensors without the need for manual assistance, which increases accuracy. AgritechMurcia platform consists of numerous agricultural technology firms that work on projects for providing ideal farming solutions. One such project developed as part of this is construction of greenhouses for organic farming.

When connected to a cloud server, the system can be remotely accessed to enable data processing. Many platforms are available for translation of data into actionable insights for farmers. Sensors can control actuators through Wi-Fi signals to perform different tasks, including regulating the heater, change light levels and more. In addition to decreasing costs,

to make it eco-friendly, solar-powered IoT sensors can be used. Emails or SMS alerts can be sent to the user regarding the current state of variables.

Ag-tech tools and techniques for crop cultivation

Precision farming is the most popular end-to-end application of the IoT in the agricultural sector. Companies like Arable, Gamaya and CropX are some of the key names in this field. Gamaya, for example, focuses on large-scale monitoring and diagnostics through hyperspectral imaging and Big Data. The business has smart solutions for soybean and sugarcane cultivation.

Using precision farming, instead of applying inputs like water, fertilisers and pesticides uniformly across the field, farmers target a specific area or even a single plant and apply inputs that optimise crop production.

Real-time weather data collected through weather stations or by different sensors includes information about various parameters like humidity, rainfall, temperature and more. Alerts are sent if weather conditions are unsuitable.

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